

Probability Density Function Quiz — Detailed Solutions and Grading

A continuous random variable Y has the probability density function

$$f(y) = \begin{cases} 2y, & 0 \leq y \leq 1, \\ 0, & \text{otherwise.} \end{cases}$$

Each question is worth **5 points** (total 20 points).

1. Verify that $f(y)$ is a valid probability density function.

A probability density function must satisfy two conditions:

1. $f(y) \geq 0$ for all y .
2. $\int_{-\infty}^{\infty} f(y) dy = 1$.

Clearly, $f(y) = 2y \geq 0$ for $0 \leq y \leq 1$.

Now check normalization:

$$\int_{-\infty}^{\infty} f(y) dy = \int_0^1 2y dy = [y^2]_0^1 = 1.$$

Thus, both conditions are satisfied.

$f(y)$ is a valid probability density function.

2. Compute the mean $E[Y]$.

$$E[Y] = \int_0^1 y \cdot 2y dy = 2 \int_0^1 y^2 dy = 2 \left[\frac{y^3}{3} \right]_0^1 = \frac{2}{3}.$$

$$E[Y] = \frac{2}{3}.$$

3. Determine the probability that Y is less than its mean.

We need $P(Y < E[Y]) = P(Y < \frac{2}{3})$.

The cumulative distribution function $F(y)$ is obtained by integrating $f(y)$:

$$F(y) = \int_0^y 2t dt = y^2, \quad 0 \leq y \leq 1.$$

Thus,

$$P(Y < \frac{2}{3}) = F(\frac{2}{3}) = (\frac{2}{3})^2 = \frac{4}{9}.$$

$$P(Y < E[Y]) = \frac{4}{9}.$$

4. Sketch (roughly) the PDF and CDF on the same axes.

Description for sketching:

- **PDF** $f(y) = 2y$
 - Starts at $(0,0)$, increases linearly to $(1,2)$.
 - Area under the curve equals 1.
 - Triangular shape.
- **CDF** $F(y) = y^2$
 - Starts at $(0,0)$ and rises smoothly (concave up) to $(1,1)$.
 - Slope equals the value of $f(y)$ at each point.

Key points to label: - At $y = 0$: $f(0) = 0$, $F(0) = 0$. - At $y = 1$: $f(1) = 2$, $F(1) = 1$. - At $y = \frac{2}{3}$ (mean): $f(\frac{2}{3}) = \frac{4}{3}$, $F(\frac{2}{3}) = \frac{4}{9}$.

Your sketch should show both curves sharing the same horizontal y -axis: - $f(y)$ as a straight line from $(0,0)$ to $(1,2)$.
- $F(y)$ as a rising quadratic curve from $(0,0)$ to $(1,1)$.

PDF: $f(y) = 2y$, CDF: $F(y) = y^2$.

Grading Guideline (5 points each)

Question	Key Steps	Points Allocation
1. Verify $f(y)$	Check non-negativity (1), set up normalization (1), compute integral (2), conclude validity (1)	5 pts
2. Mean $E[Y]$	Set up $E[Y] = \int yf(y) dy$ (1), integrate correctly (3), final result $\frac{2}{3}$ (1)	5 pts
3. Probability $P(Y < E[Y])$	Express with CDF (1), compute $F(y) = y^2$ (2), evaluate $F(2/3)$ (1), final result $\frac{4}{9}$ (1)	5 pts
4. Sketch PDF and CDF	PDF shape correct (2), CDF shape correct (2), labels/key points clear (1)	5 pts

Partial credit rules: - Arithmetic or algebraic slips: deduct 1–2 pts. - Forgetting to normalize or omitting domain boundaries: deduct up to 2 pts. - Missing CDF derivation or incorrect sketch labels: partial deductions as appropriate.

Final Answers Summary

$$f(y) = 2y, \quad 0 \leq y \leq 1,$$

$$E[Y] = \frac{2}{3},$$

$$P(Y < E[Y]) = \frac{4}{9},$$

$$F(y) = y^2,$$

PDF shape: linear $(0,0) \rightarrow (1,2)$, CDF shape: quadratic $(0,0) \rightarrow (1,1)$.